

AE-533: Liquid phase calibration in Argon of normal range H2 sensor

Date: 2025-08-18

Tags: Test AE Calibration H2 in-situ
Unisense electrochemical H2 detection
H2/O2 reactor

Category: Prep work

Status: Done

Created by: Alexander Eith

Literature/reference experiments

Literature	/
Reproduction	/
Similar experiments	H2 Evolution - AE-531: Liquid phase calibration of normal range H2 sensor

Reagents

Name	CAS Number / Experiment Number	Inventory number	pressure [bar]	Volume [ml]	mass [g]	Concentration [M]
hydrogen	1333-74-0	/	approx. 1	2 ballons	/	/
Milli-Q water	/	/	/	25	/	/

Procedure/observations

Date	Time	Step	Observations	Pictures
18.08		The liquid phase calibration according to Protocol - Liquid phase calibration of H2 UniAmp sensor was started, with some deviations with Equipment - H2 UniAmp Sensor - Normal range - 2.1 x 80 mm needle . The steps taken are shortly presented below with timestamps and deviations and observations.	/	/
	09:05	The setup was built using the H2/O2 reactor, the PT1000 connected to the Equipment - H2 UniAmp Single Channel System , and a Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel (Firesting 2) with the Equipment - Robust probe for liquid O2 measurement and a PT100. The PT100 was placed outside the reactor in the advanced irradiation setup. PTFE tubing was connected to the valve, which was connected to an argon line. The H2 sensor was placed outside of the reactor. The still open GL connections were closed with GL caps. The reactor was filled with water (25 mL) using a 25 mL graduated cylinder.	for picture see 09:34	/
	09:23	The oxygen log on the firesting was started	AE-533-1-02	/
	09:34	The degassing of the H2/O2 reactor with argon was started	/	setup argon sparging.jpg

	09:35	The hydrogen log was started	AE-533-1-H2-logger1	/
	09:41	In a 5 mL rbf water (approx. 5 mL) was added and a hydrogen ballon was connected, starting the hydrogen bubbling	/	setup H2 bubbling.jpg top left rbf for H2 bubbling.jpg
	09:52	The hydrogen log was stopped	/	/
	09:53	The H2 sensor was moved to the H2/O2 reactor in Argon counterflow	/	setup for calibration.jpg setup for calibration side view.jpg
	09:54	The hydrogen log was started	AE-533-1-H2-logger2	/
	09:55	The degassing of the H2/O2 reactor was stopped	/	/
	09:59	The 0 ppm H2 saturated water point was taken	0.7 mV	/
	10:00	Hydrogen saturated water was added (0.25 mL) to the reactor using 1 mL syringe	/	/
	10:01	The 9901 ppm H2 saturated water point was taken	10 mV	/
	10:01	Hydrogen saturated water was added (1.0 mL) to the reactor using 1 mL syringe	/	/
	10:02	The 47619 ppm calibration point was taken	38.5 mV	/
	10:03	A hydrogen ballon was attached to the reactor and H2 bubbling was started. After approx 5 min the ballon was exchanged by a new Hydrogen ballon	/	setup H2 bubbling in reactor close up.jpg setup H2 bubbling in reactor.jpg
	10:17	The calibration point with 1.000.000 ppm H2 was taken	1108 mV	/
	10:18	The H2 bubbling of the reactor was stopped	/	/
	10:19	The reactor was opened to the enviroment	/	/
		The calibration was saved with all data points	AE-533-calib-all_data.bmp	/
		The calibration was saved without 1.000.000 ppm points	AE-533-calib_without_1000000ppm.bmp	/
		The calibration was saved without 47619 ppm points	AE-533-calbi_without 47619ppm.bmp	/
		The calibration was saved without 47619 and 1.000.000 ppm points	AE-533-calib_wothout_1000000_and_47619ppm.bmp	/
	10:20	The H2/O2 reactor as flushed with argon	/	/
	10:20	The logging of O2 and H2 was stopped	/	/

Analysis

Date	Time	Sample name	Analysis method	Analytical device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
18.08	09:23	AE-533-1-H2-logger1	electrochemical H2 detection	Equipment - H2 UniAmp Sensor - Normal range - 2.1 x 80 mm needle	water	AE-533-1-H2.ulong	AE-533-1-H2-logger1.png	H2 Evolution - AE-531: Liquid phase calibration of normal range H2 sensor	Stable after approx. 10 min, signal of 0.7 mV
18.08	09:25	AE-533-1-H2-logger1	electrochemical H2 detection	Equipment - H2 UniAmp Sensor - Normal range - 2.1 x 80 mm needle	water	AE-533-1-H2.ulong	AE-533-1-H2-logger-2.png AE-533-calib-all_data.bmp AE-533-calib_without_1000000ppm.bmp AE-533-calib_without_47619ppm.bmp AE-533-calib_without_1000000_and_47619ppm.bmp	H2 Evolution - AE-531: Liquid phase calibration of normal range H2 sensor	all data: slope: 1.436, Intercept: -4.822, R2: 1.000 w/o 1.000.000 ppm: slope 1.001, intercept: 1.456, R2: 0.999 w/o 47619 ppm: slope: 1.433, intercept: -0.111, R2: 1.000 w/o 1.000.000 and 47619 ppm: slope: 1.174, intercept: 0.873, R2: 1.000
18.08	09:54	AE-533-1-O2	Optical O2 detection	Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel	water	AE-533-1-O2.txt	AE-533-1-O2.png	/	Degassing worked till 0.6 μ M, increase to 2 μ M upon stopping, decrease to approx. 1 μ M upon H2 bubbling

Results

Worked nicely, similar slope obtained compared to AE-531. Good linearity obtained.

O2 measurement during H2 bubbling shows only minor amounts of H2 in balloon and also in H2 saturated water low amount of O2, since no spike upon addition of H2 saturated water was observed

Linked experiments

H2 Evolution - [AE-492: Liquid phase calibration of H2 sensor](#)

H2 Evolution - [AE-498: Liquid phase calibration of H2 sensor](#)

H2 Evolution - [AE-504: Liquid phase calibration of H2 sensor I](#)

H2 Evolution - [AE-528: Initial test of H2 sensor \(normal range\)](#)

H2 Evolution - [AE-530: Gas phase calibration of Normal range H2 sensor](#)

H2 Evolution - [AE-531: Liquid phase calibration of normal range H2 sensor](#)

Prep work - [AE-477: Testing of in-situ electrochemical hydrogen sensor and test H2 evolution reaction with RhCrO:Al:SrTiO3](#)

Linked resources

Equipment - [Firesting Fiber-Optic Oxygen Meter 2 Channel \(Firesting 2\) - CEEC II E004 C9](#)

Equipment - [Robust probe for liquid O2 measurement-1](#)

Equipment - [H2 UniAmp Single Channel System](#)

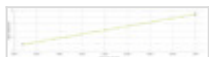
Equipment - [H2 UniAmp Sensor - Normal range - 2.1 x 80 mm needle - CEEC II E002 C2](#)

General Protocol - [Liquid phase calibration of H2 UniAmp sensor](#)

Attached files

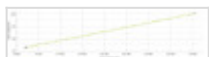
AE-533-calib_wothout_1000000_and_47619ppm.bmp

sha256: 720568f2e7c937c7cdc0fef70af0f93d975409af5bec8bfd1a6170d8dbb05145



AE-533-calbi_wothout 47619ppm.bmp

sha256: 5e54a1a25068a5737646a0b937558fc817a7c6fa2571310b8bb58e7a5a56043b



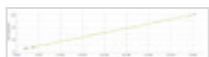
AE-533-calib_wothout_1000000ppm.bmp

sha256: 844245eb9de91dfa0d08b1e5a505988e7f2400051e452470dfa5b829bf54e744



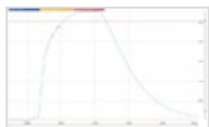
AE-533-calib-all_data.bmp

sha256: edf87db1142ae28bc5acbb7d0c4f3d01ef257580ec8547d2b6ad8611bbb6e843



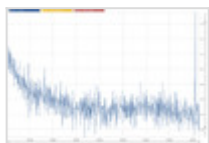
AE-533-1-H2-logger-2.png

sha256: 61a3d7c37f771d3f500fb515c2d992ee8495bcf685f42faf081be953cb84c9c8



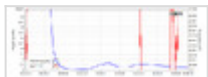
AE-533-1-H2-logger1.png

sha256: 47dc7c419d98ae590c27d77bbf9cb47adac436580c2b49e6ef1dd7fb6c370b03



AE-533-1-O2.png

sha256: f65883aa2cd9b759d3f858a53146be1612d8a3d89665f8aac7a66a27be632af0



AE-533-1-O2.txt

sha256: f43e9484d04e2b0b474ef49255cc4bd8d197afdd6aaaa9ae746f49700a154e2c

AE-533-1-H2.ulog

sha256: c8bea1354596069ab4ac3ee6c46248116a56aa51f954c8b1f58013f778849207

setup-H2-bubbling-in-reactor.jpg

sha256: b0c59a9807ca4a53af75f109d0e89b4d9d0e5a6815fc01e080f9a87a426d1b6d



setup-H2-bubbling-in-reactor-close-up.jpg

sha256: 0d510a84ac58f1bbca2b5c25c1fdb5d4ea4f4be100bf70e3480d84f3f155a196



setup-H2-bubbling.jpg

sha256: 7528700b83c4902318a2d1c96f18dcf4f28c13ad0eb7a2c3b37e3e45c786023f



setup-for-calibration-side-view.jpg

sha256: 9287d1c61f92cab1af377e0eecf83dc36cec40193a90a965c0e46ce281aae976



setup-for-calibration.jpg

sha256: d0c2b2cf77dfad9b9d15cf56e730f7ccc3207ba7119988ed2f94cd199b8c6b52



setup-ar-sparging.jpg

sha256: a5d06faf3316a5c023a0eea1e3a0a2f6deacf7b1dc058d91c32b03b80d8420a2



top-left-rbf-for-H2-bubbling.jpg

sha256: fade72658391c2d9f3df90d3500e9fb815db8ba115f5791b584e7aacb413b588



Unique eLabID: 20250818-40022713952fe4a8f1fe4da70d0ca4d9a1c0ee30

Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=2708>